Mindful Muslim- Salah Habit Tracker for IIUM Student

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***Abstract*—MindfulMuslim is a mobile application developed to support Muslims in maintaining regular and disciplined salah practices, particularly targeting the student community at International Islamic University Malaysia (IIUM). The app addresses common challenges such as inconsistent prayer habits and lack of motivation by offering features for tracking prayer performance, categorizing prayer statuses, and logging jamaah prayers.By fostering community engagement and accountability, MindfulMuslim aims to enhance users' spiritual discipline and provide a practical, user-friendly solution for improving prayer consistency. This report outlines the system's objectives, design, and development process, detailing how the app meets the needs of its target users.**

Keywords—salah,student,prayer,habits,track,spiritual, motivation.

I. INTRODUCTION

MindfulMuslim is a mobile app designed to help Muslims maintain consistent salah practices, focusing on the IIUM student community. Salah, a fundamental pillar of Islam, can be challenging to perform regularly due to busy schedules and other distractions. The app addresses this by providing features like prayer tracking, categorized status options, jamaah logging, By fostering a supportive and community-driven environment, MindfulMuslim aims to enhance spiritual discipline and strengthen users' connection to their faith.

A. Background of the Problem

In today's busy and distracting environment, many students at International Islamic University Malaysia (IIUM) struggle to consistently perform their five daily salah, a key pillar of Islam highlighted in Surah Al-Baqarah (2:43). This issue is worsened by the lack of a centralized platform for tracking prayer performance, limited community engagement for mutual motivation, and inadequate tools to categorize prayer statuses such as on time, late, in jamaah, or missed. Without a supportive system, students may lose track of their prayers, hindering spiritual growth and discipline

B. Problem Statement

Muslims are obligated to perform five daily prayers, but many struggle to consistently fulfill this due to challenges such as busy schedules, lack of reminders, and gaps in spiritual discipline. Existing habit trackers fail to meet the unique needs of Muslims, lacking features like prayer timing adherence, progress analysis, and spiritual motivation. This highlights a gap in the market for a dedicated, user-friendly, tech-driven solution to help Muslims track and improve their prayer habits effectively.

C. Objectives

The Salah Tracker app is designed to help IIUM students strengthen their prayer consistency by addressing challenges that often disrupt regular salah practices. It seeks to foster a supportive environment by promoting community engagement and mutual encouragement among peers, enabling students to motivate each other in maintaining consistent prayer habits. Additionally, the app provides practical and accessible tools to support students in fulfilling their religious obligations with greater ease and discipline. Through these efforts, the Salah Tracker aims to contribute to the spiritual growth and well-being of its users.

D. Scope

1) Log and Track:The Salah Tracker system focuses on enhancing prayer consistency among IIUM students by providing features to log and track salah performed at various campus facilities.

2) Target User: IIUM Students including Undergraduate and postgraduate students across all kulliyyahs who aim to track their salah performance, engage in challenges, and foster prayer discipline.

3) Platform Used are LucidChart , Figma , React Native , Expo , Firebase and VScode.

II. Literature Review

A. Introduction

The primary purpose of this literature review is to gain an understanding of four existing systems through an analysis that highlights both their similarities and differences. This serves as an input for identifying the features, design, and functionalities to be implemented in the MindfulMuslim Salah Tracker mobile app and the improvements that need to be introduced.

B. Existing Systems

The four existing systems under review are Everyday Muslim: Prayer Times [7], Daily Deeds: Prayer Tracker [8], eEbadat: Islamic Habit Tracker [9], and Hudi: Build Jannah Habits [10]. A thorough review of these mobile applications has helped to identify the necessary features required in a Salah tracker system. The similarities and differences among the systems are presented in Table 1 for easy comparison with Mindful Muslim

Table 1 - Comparison of system features

App	Weekly /Monthly Records	Challenge	Prayer Hadith	Reminder
Mindful Muslim	✅	✅	✅	✅
Daily Deeds	✅	❌	❌	✅
Everyday Muslim	❌	✅	❌	❌
eEbadat	❌	❌	❌	✅
Hudi	❌	❌	✅	✅

C. System Adaption

Based on the information from Table 1, this review outlines how features from the reviewed systems were integrated into the MindfulMuslim Salah Tracker. Key features, such as the Weekly/Monthly Prayer Records from Daily Deeds, allow users to track their prayer habits over time, providing an overview of their progress. The Challenge Concept from Everyday Muslim encourages users to improve their prayers by committing to specific durations, like a week, promoting consistency in Salah. Additionally, the Educational Content from Hudi, featuring motivational Hadith, helps users

D. Improvement Introduced

Several improvements have been introduced in the MindfulMuslim Salah Tracker to enhance the user experience. First, the app offers an Intuitive Weekly/Monthly View, building on the percentage summary from the Daily Deeds app. This feature includes insights like the "Total Best Day," highlighting days with on-time or Jamaah prayers and no missed or late prayers. The app also introduces Social Features for Community Engagement, inspired by Everyday Muslim, with an "Invite Friend" option to share weekly records and foster accountability. Additionally, the app allows users to log Jamaah attendance manually if they are at a registered Jamaah venue during early prayer times, making it easier to track communal worship participation. Finally, Personalized Motivational Content will be use to encourage consistency, sending Hadith and reminders if Salah is miss consecutively.

III. METHODOLOGY

A. Development Approach

The waterfall methodology is a linear and sequential documentation focused approach where each phase introduction, literature review, methodology, design, and prototyping is completed before moving to the next. It is ideal for projects with well-defined, stable requirements, like the Mindful Muslim Salah Tracker, where changes are minimal. This method ensures thorough planning and documentation at each stage, aligned with the seminar-based structure. Weekly seminars provide clear guidelines for each phase, ensuring systematic progress while meeting both academic and functional objectives

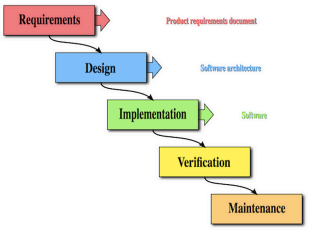


Figure 1 - Waterfall Methodology

For coding, testing, and development of the Mindful Muslim Salah Tracker, Agile methodology suits best. Its iterative, flexible approach supports continuous user feedback, particularly for Salah tracking, reminders, and analytics. Time-boxed sprints enable cyclic development, testing, and refinement, ensuring high-quality, user-centric features. This hybrid blends Waterfall's planning with Agile's adaptability for efficient development.

B. Use Case Diagram

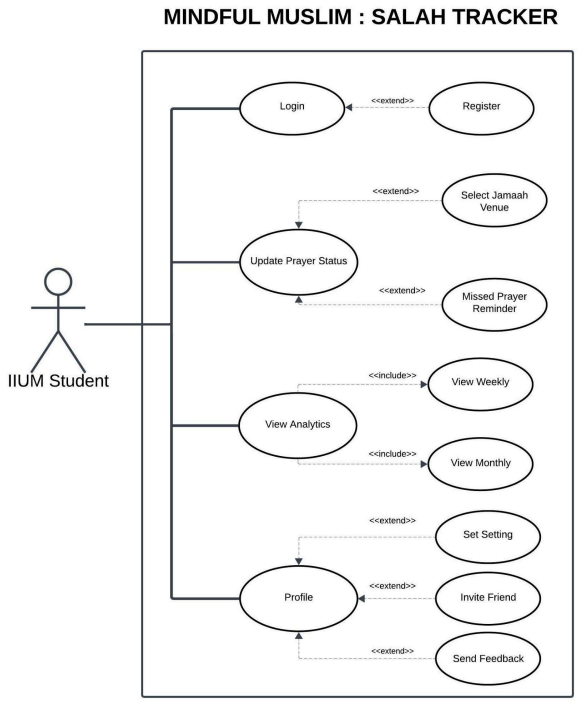


Figure 2 - Use Case Diagram

This use case diagram outlines the functional requirements and user interactions within the Mindful Muslim app. The diagram includes one primary actor, IIUM Students, who interact with various system features to track and update their prayer habits mindfully. The first use case is Login, which allows users to access the app, with an Extend to Register if the user is not yet registered. Once logged in, students can Update Prayer Status. This use case also triggers a Reminder with Hadith and course of action for each missed prayers. The View Analytics use case enables users to track prayer progress with options for Monthly and Weekly View. Another feature is Track with Friend, allowing users to compare progress with friends. Additionally, students can Edit Profile, including an Extend to Settings for profile-related changes. Finally, users can Send Feedback to improve the app. The relationships between use cases, using Include and Extend, structure the app's functionality efficiently, helping Mindful Muslims stay consistent and engaged in their spiritual practices.

C. Database Design

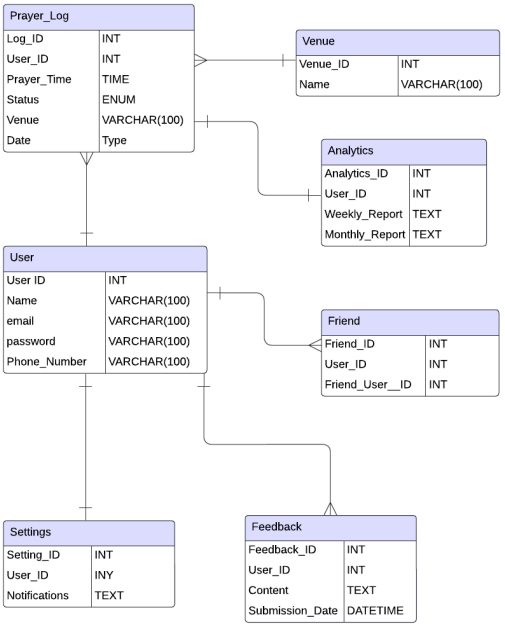


Figure 3 - Entity Relationship Diagram

Figure 3 illustrates the Entity Relationship Diagram (ERD) for the Mindful Muslim Salah Tracker App, which involves seven entities.

1. *User:* This table stores user information, including the unique User_ID, Name, Email, Password, and Phone_Number.
2. *Prayer_Log:* This table tracks individual prayer records for users. It helps users monitor their prayer habits.
3. *Venue:* This table contains information about prayer locations,. It is linked to the Prayer_Log entity to record where prayers are performed.
4. *Analytics:* This table provides insights into a user's prayer performance.
5. *Friend:* This table manages connections between users, enabling peer engagement and accountability.
6. *Settings:* This table handles user-specific app settings.
7. *Feedback:* This table records feedback submitted by users.

IV. RESULT

A. System Integration

The MindfulMuslim app integrates Firebase to enable real-time data synchronization for key features such as prayer records, user authentication, masjid locations, and analytics. Firebase services, including Firestore and Authentication, support efficient and scalable backend operations. The app is developed using React Native with TypeScript, providing strong typing for improved code quality and maintainability. Development is carried out in Visual Studio Code, with Expo used to streamline the development, testing, and deployment process. This setup ensures rapid, organized, and cross-platform mobile development.

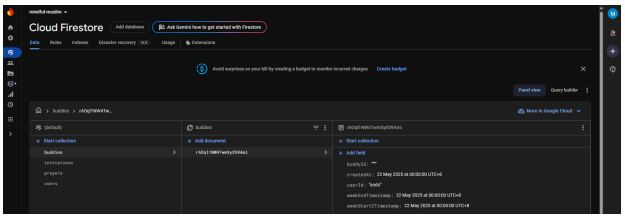


Figure 5 - Database page of Firebase

Tool/Framework	Logo	Purpose
Visual Studio Code		Integrated Development Environment (IDE) for coding
Firebase		Real-time database, authentication, analytics
React Native		mobile app development framework
TypeScript	TS	Programming language used
Expo	Expo	React Native tooling for faster development

Table 2 - Tools and Frameworks Used in App Development

The application frontend is built using **React Native**, a JavaScript framework for developing cross-platform mobile apps. In our project, we extended this by using **TypeScript**, a typed superset of JavaScript that offers better tooling, readability, and error-checking, which is especially beneficial for a growing codebase for this app.

B. System Output

The MindFul Muslim system output below highlights the main functionalities of MindFul Muslim Mobile App Application which enables users to track their salah activity with reminders, tracking database and tracking with friends functions

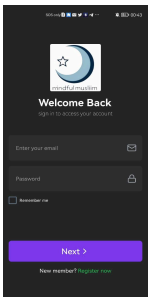


Figure 6 - log in page

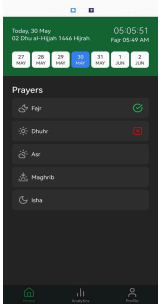


Figure 7 - Home Page

Figure 6 shows the log in page where user key in their account details. Next, Figure 7 shows the home page where the app displays the prayer status with the dates and also the countdown to the next prayer time

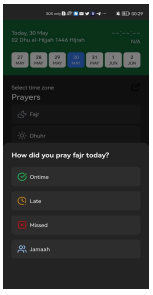


Figure 8 - Prayer Status page

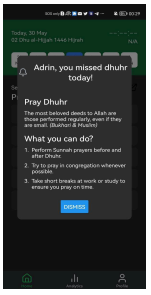
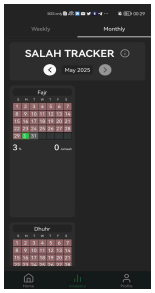


Figure 9 - Reminder Page

Figure 8 shows prayer status page where user key in their prayer status.Next,Figure 9 shows reminder page where the user will receive a reminder if the user keyed in the not prayed status.



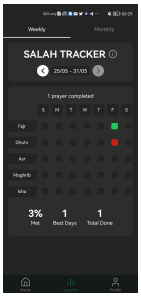
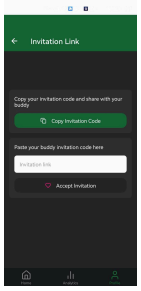


Figure 10 - Monthly Analytics *Figure 11 - Weekly Analytics*

Figure 10 shows Monthly analytics page where the page shows the prayer status in a monthly view. Next, Figure 11 shows the analytics of prayer status in a weekly view.



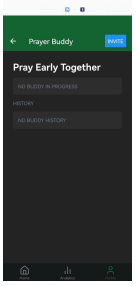


Figure 12 - Invite Friend *Figure 13 - Prayer buddy*

Figure 12 shows invite friend page where user can copy a WhatsApp link to invite friend to track their prayer status together.. Next Figure 13 shows the prayer busy status if the user has a prayer buddy where they can track prayer together.

C. System Flow

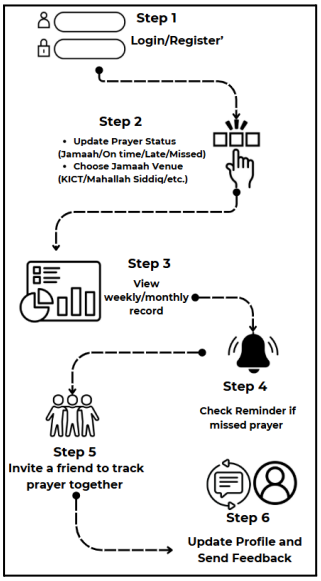


Figure 14 - Mindful Muslim App Flow

The system flow outlines the key user journey within the Mindful Muslim app. Users begin by logging in or registering, then update their prayer status and select a Jamaah venue. Prayer times are automatically set based on the user's location using the JAKIM official system . Users can view weekly or monthly prayer records and receive reminders for missed prayers. The app also allows inviting friends for shared tracking and provides options to update the profile and submit feedback. This flow ensures a smooth, personalized, and engaging user experience.

D. System Testing

The User Acceptance Test (UAT) was conducted in person. The first session was held with the project supervisor, Dr. Mimi Liza. The second session involved the target users of the mobile app, which are IIUM students. Four students participated in the second session. Those student are Fatin , a student from the Kulliyyah of Communication Technology; Ahmed, a student from the Kulliyyah of Islamic Revealed Knowledge; Luqman, a first-year student from the Kulliyyah of Economics; and Fikri, a postgraduate student from the Kulliyyah of Human Sciences.

E. Testing Feedback

Based on UAT feedback, the following enhancements were suggested to improve the Mindful Muslim app:

1. Wearable Integration
   Support for devices like Wear OS to allow prayer updates and notifications via smartwatch.
2. Language Options
   Add support for multiple languages such as Malay and Arabic for better accessibility.
3. Quranic Tracker
   Include a feature to track Quran reading progress alongside prayer tracking.
4. Motivational Features
   Introduce streaks, badges, and weekly goals to encourage consistent prayer habits.
5. Calendar Sync
   Enable syncing of prayer records with Google Calendar or other personal calendars.

V. DISCUSSION

A. Interpretation of Results

MindfulMuslim successfully achieved its objectives by introducing a unique feature that distinguishes it from other Islamic habit tracking apps a structured system for tracking daily prayers and spiritual habits tailored specifically for Muslims. This feature enhances the user experience by promoting consistency in worship and offering gentle reminders that align with Islamic values. The system demonstrated strong effectiveness in fulfilling its goals, as it supports the personal development of users, aligns with the spiritual and lifestyle needs of the Muslim community, and encourages mindfulness and discipline. Feedback from initial users highlighted the app's user-friendly interface, its meaningful and culturally relevant features, and its role in helping users develop a deeper connection with their faith, affirming its value as a purposeful lifestyle tool.

B. Limitation

MindfulMuslim's current limitations include the high cost associated with integrating the Google Maps API, which impacts the scalability of location-based features such as finding nearby mosques or prayer areas. Additionally, the app is currently developed exclusively for Android devices due to the complexities and higher development requirements for iOS, which limits accessibility for iPhone users. These limitations highlight important areas for future enhancements to improve the overall user experience and scalability of the platform.

C. Conclusion

In conclusion, MindfulMuslim has introduced impactful and faith-centered features that support the daily spiritual routines and personal growth of Muslims. Its unique approach to habit tracking rooted in Islamic practices sets it apart from generic wellness apps. The platform has proven effective in encouraging consistent prayer, mindfulness, and self-discipline, aligning well with users' spiritual goals. Planned expansions, including iOS and Wear OS support, as well as more cost-efficient and scalable solutions, are expected to further strengthen the app's reach and impact. MindfulMuslim stands as a promising tool in supporting a purposeful and spiritually balanced lifestyle.

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